

CHAPTER 9 OF 13 • PART FOUR

The Reliability Kit

Failure is the environment, not the exception. Redundancy only helps when the copies can fail apart.



Thirteen chapters, gathered into six movements.

01 A Tuesday — one engineer, wake to sleep

02 The words — naming what you just watched

03 Protagonist and antagonist — where a design starts

04 Solution types — which answer is wanted

→ **Six kits — data, compute, network, scale, reliability, security**

06 Two designs, then the map back — Instagram, a parking app, and what you keep

WHERE THIS SITS

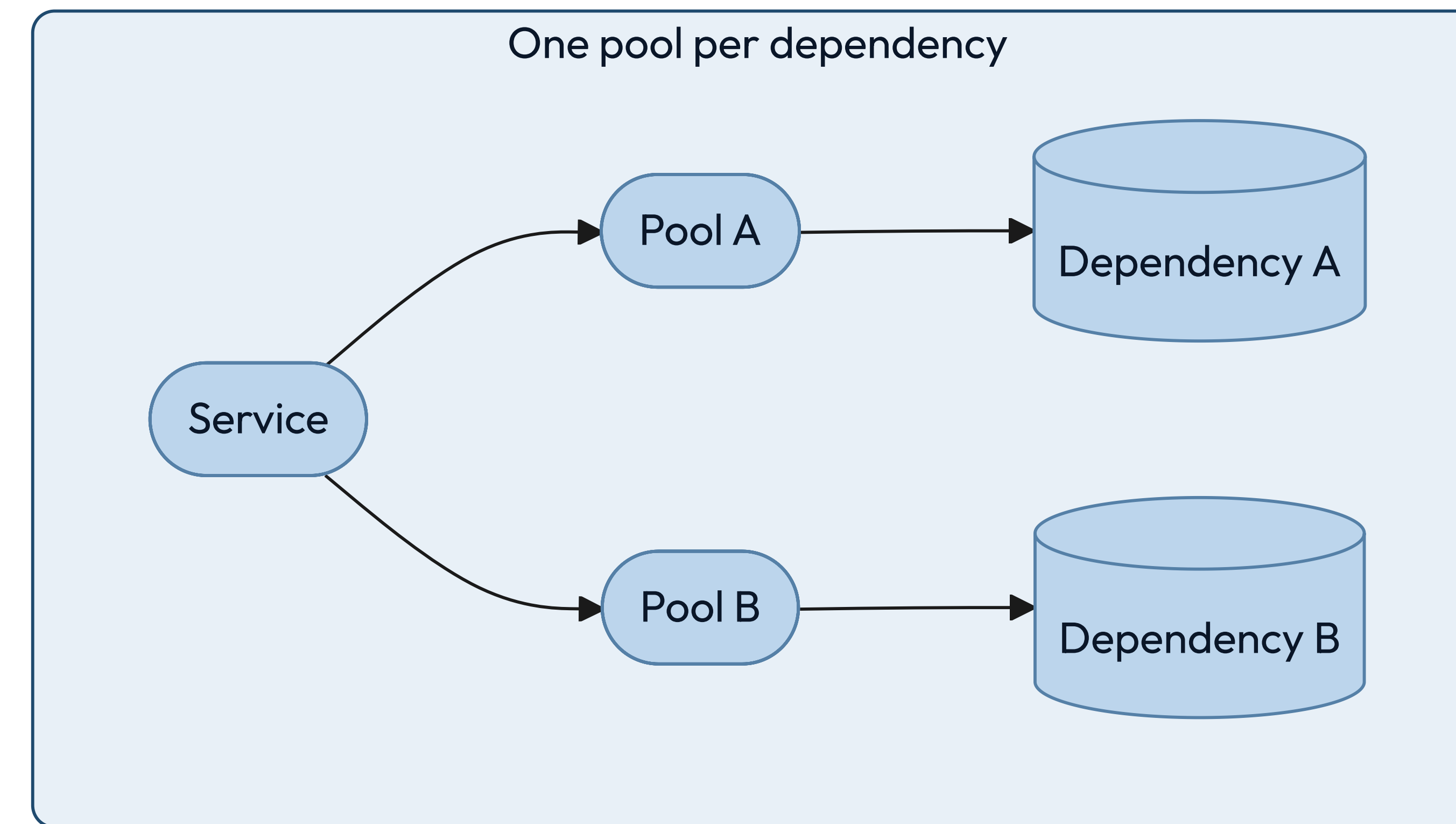
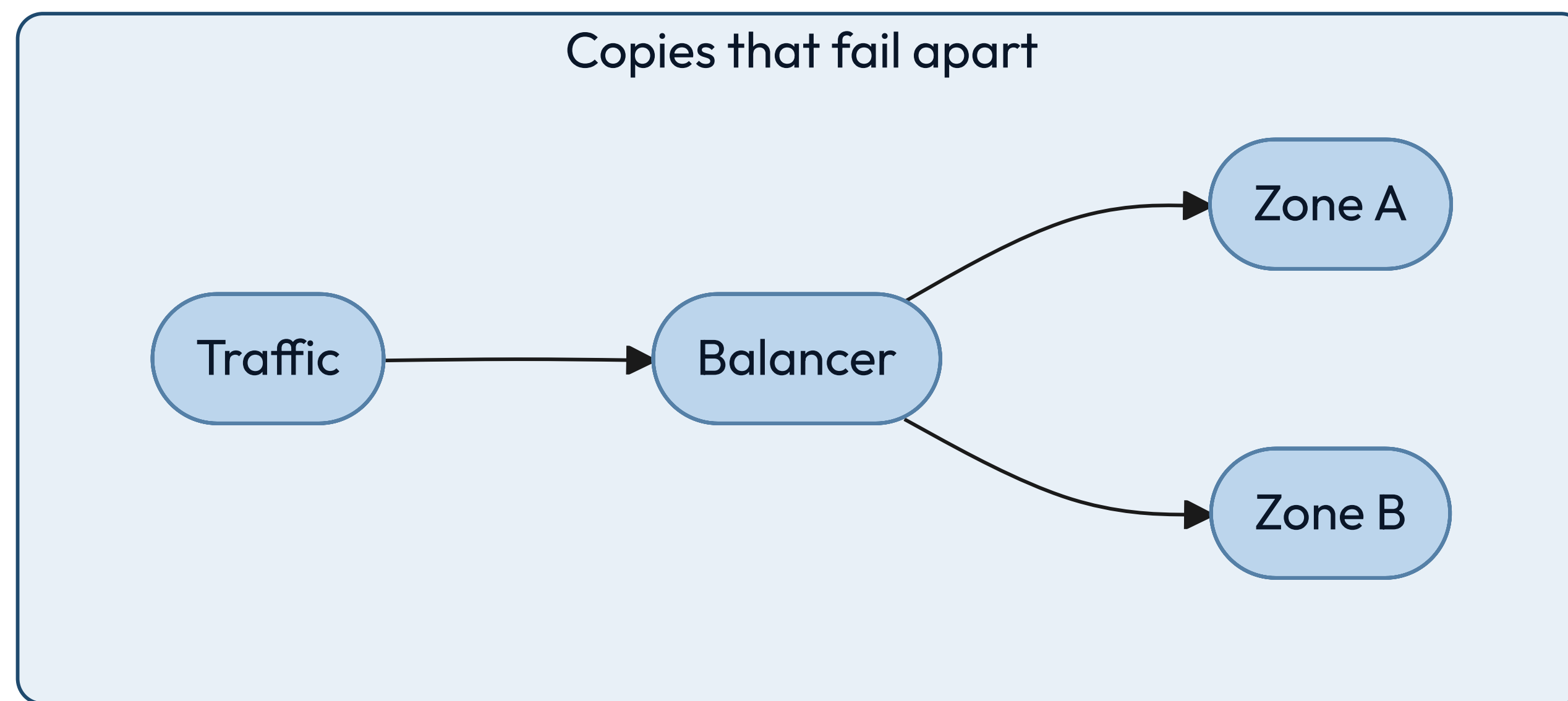
The scale kit asked for admission control. This chapter is where it lives.

Chapter eight's third invariant asked you to shed load before the system collapses — a reliability pattern doing scale's work. The two kits differ in the question they ask: chapter eight asks what happens when there is more of everything, this one asks what happens when one part of it stops. It assumes the four scaling moves and the tail latency from chapter eight.

RELIABILITY

**Failure is the environment,
not the exception.**

Two of the four patterns put a wall between what failed and what has not.

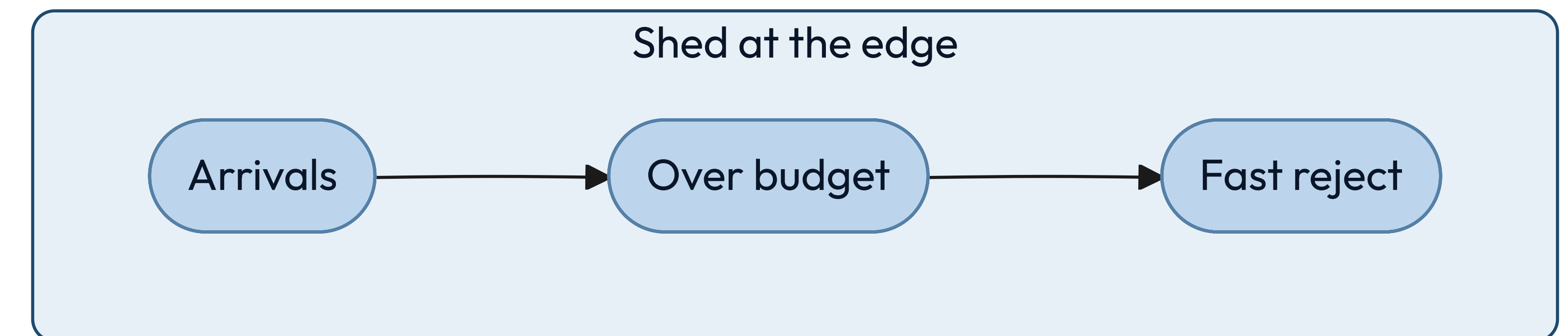
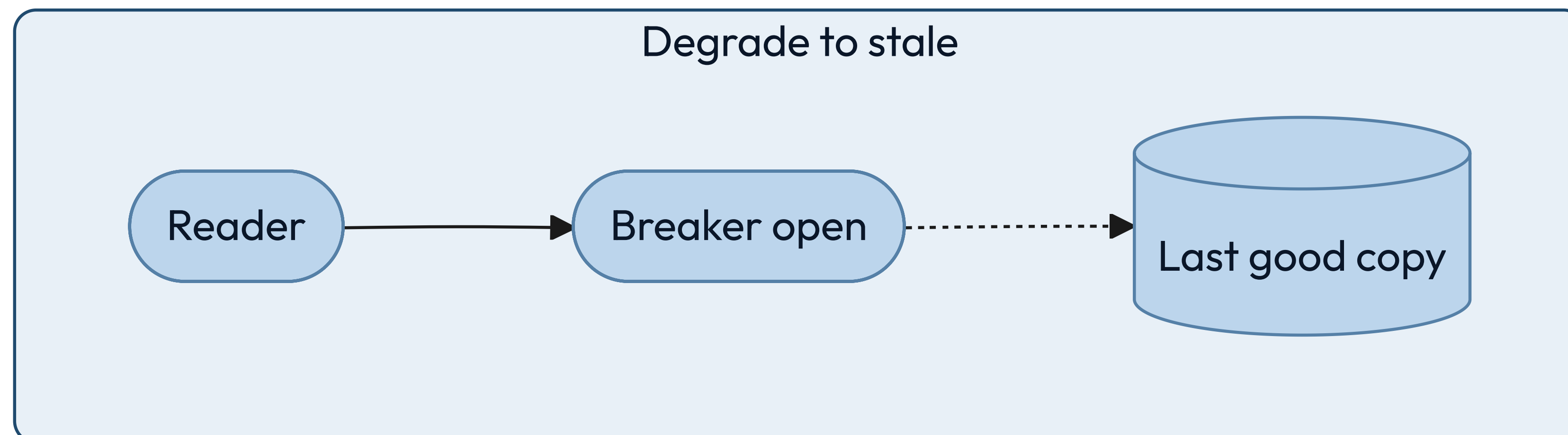


KEY INSIGHT

A wall only works if the two sides do not share the thing that broke.

★ The scale kit's third invariant asked for admission control — shed load before the system collapses. That is a reliability pattern doing scale's work, and it is on the next slide. The two kits differ in the question: scale asks what happens when there is more of everything, reliability asks what happens when one part of it stops.

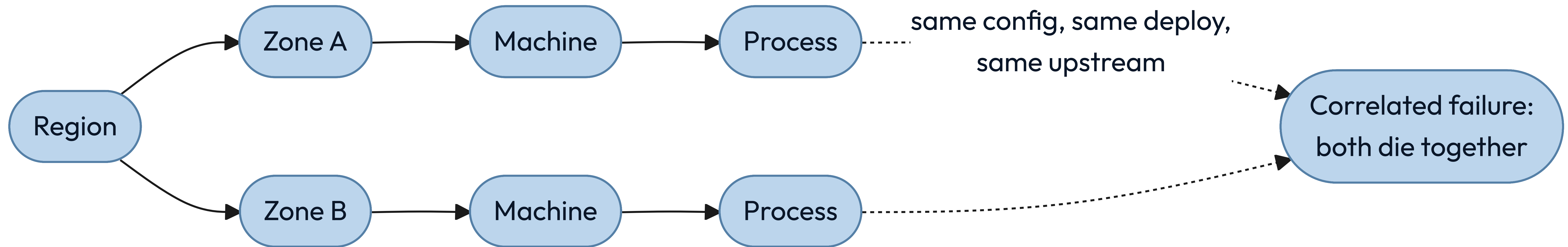
The other two answer anyway, rather than make the caller wait.



KEY INSIGHT

Stale is an answer and refused is an answer. Timing out is neither.

Redundancy only helps when the copies can fail apart.



KEY INSIGHT

Two copies of one mistake are still one copy.

A slow dependency takes the whole building unless something bounds the wait.

STEP 01

Timeout

Bound the wait, so a slow callee cannot hold your resources.



STEP 02

Retry with backoff

Recover from a blip, with a budget and jitter. Retries without them are a reinforcing loop.



STEP 03

Circuit breaker

Stop calling something that is clearly down, and let it recover.



STEP 04

Bulkhead

Give each dependency its own pool, so one queue cannot drain them all.

Four words people use interchangeably decide whether you may ship this week.

01 SLI

The measurement: the share of requests served under 300 milliseconds.

02 SLO

Your internal target for it, such as 99.9 percent over 28 days.

03 SLA

The external promise, with money attached, and always looser than the SLO.

04 Budget

What the objective lets you spend. Exhausted, feature work stops.

An availability target is a budget that shrinks fast.

01	99 percent	3.65 days per year
02	99.9 percent	8.8 hours per year
03	99.99 percent	53 minutes per year
04	99.999 percent	5.3 minutes per year

READING THAT TABLE

Past three nines the humans stop being fast enough.

Fifty-three minutes a year is about one incident with a page, a login, a look at a dashboard and a decision. So four nines does not mean a faster on-call engineer. It means automatic failover and automatic rollback, because a person is no longer in the loop.

Each nine roughly multiplies the cost. Pick the number the business actually needs, and write down what you are choosing not to buy.

Each signal answers a different question, and none of them replaces another.

Metrics

Cheap, aggregate, always on.
They tell you that something is wrong.

Logs

Detailed, expensive, one event at a time. They tell you what happened in one case.

Traces

One request across every service.
They tell you where the time actually went.

KEY INSIGHT

If you cannot answer "which dependency is slow" inside a minute, you have metrics, not observability.

RELIABILITY KIT • DEGRADATION

A well-designed system gets worse in an order somebody chose.

Under pressure something has to give. Either you decided in advance which features degrade, or the system decides at random and drops the one that takes money.

THE RULE

Every feature has a stated tier, and the lowest tier fails first by design.

READ PATHS OUTLIVE WRITE PATHS

Serving something slightly stale beats serving an error page, for almost every product.

THE FALLBACK IS EXERCISED

An untested degraded mode is untested code running in your worst hour.

Production earns trust one of these at a time.

01

Every dependency has a defined failure behavior

Written down: degrade, queue, or fail fast. Three teams found out at 09:05 that nobody had.

02

Redundant copies fail independently

Different zones, different deploys, different upstreams. Otherwise it is one copy.

03

Recovery is practiced

Restores, failovers and rollbacks happen on a schedule, not for the first time at 3am.

04

The system tells you before a user does

Alerts fire on the indicator, never on the complaint.

YOUR TURN

Three dependencies go slow rather than down. Say what the page does.

Your service renders a product page inside a 300-millisecond budget, calling a price service, a recommendations service and a reviews service in parallel. Each in turn starts answering in two seconds instead of forty milliseconds.

For each, say what the page shows and which containment pattern makes it do that. Then turn the page.

ONE ANSWER

Only one of the three is allowed to fail the page.

- 01

Price

Fail it. A product page with no price is wrong, not degraded, so it is the top tier: a timeout, then a fast error rather than a spinner.
-
- 02

Recommendations

Drop the strip and render. A breaker opens after the first few slow calls, so every other page does not pay two seconds to learn the same thing.
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- 03

Reviews

Serve the last good copy from behind a breaker of its own. Read paths outlive write paths, and stale is an answer where a timeout is not.

CHAPTER 9 OF 13 • HOW TO THINK ABOUT SYSTEMS

Copies only help when they fail apart.

*Chapter ten asks the same question about
credentials, and it starts from a harder assumption
than this one: the boundary is already crossed.*



Glossary

S

TERM	DEFINITION
SLA	The external promise, with money attached, always looser than the objective.
SLI	The measurement: the share of requests served inside your target.
SLO	Your internal target for that measurement, over a stated window.